RELATIVE DIMENSION BOUNDS FOR FLAT MAPS PROPAGATE FROM A DENSE OPEN OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8590-8601 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8603-8622, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One of the chapter's dimension-propagation family; the hypothesis split (finite presentation or quasi-compact open) is retained exactly.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Scheme-theoretic density of quasi-compact dense opens under reduction (Morphisms of Spaces), etale-local character of relative dimension (same chapter's definition section), and the schemes propagation theorem (More on Flatness, lemma-flat-finite-presentation-dimension-over-dense-open).

Source reference. Original lemma 081L: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8622–8622. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 088G: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8622–8622. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8622–8622. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0831: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8622–8622. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and locally of finite type. Let $V \subset Y$ be an open subspace such that $|V| \subset |Y|$ is dense and such that $X_V \to V$ has relative dimension $\leq d$. If also either*

(1) *f is locally of finite presentation, or*
(2) *$V \to Y$ is quasi-compact,*

then $f : X \to Y$ has relative dimension $\leq d$.

Proof. We may replace Y by its reduction, hence we may assume Y is reduced. Then V is scheme theoretically dense in Y, see Morphisms of Spaces, Lemma 088G. By definition the property of having relative dimension $\leq d$ can be checked on an étale covering, see Morphisms of Spaces, Sections 04NH. Thus it suffices to prove f has relative dimension $\leq d$ after replacing X by a scheme surjective and étale over X. Similarly, we can replace Y by a scheme surjective and étale and over Y. The inverse image of V in this scheme is scheme theoretically dense, see Morphisms of Spaces, Section 0831. Since a scheme theoretically dense open of a scheme is in particular dense, we reduce to the case of schemes which is More on Flatness, Lemma 081L. □